

Imperfect repair in degradation processes: A Kijima-type approach

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Abstract

We consider a Wiener process with linear drift for degradation modeling. Regularly, maintenance actions are carried out which produce a reduction of the degradation level. In this paper, we consider the influence of such maintenance actions to the further development of the degradation process and the resulting lifetime distribution. A connection between virtual age in Kijima-type models and degradation level in the underlying degradation process is developed. Furthermore, estimators for the process parameters as well as for the degree of repair are developed.

KEYWORDS

degradation process, imperfect repair, parameter estimation, Wiener process

1 | INTRODUCTION

Investigating the lifetime of products or biological organisms that are affected by degradation, damage, stress, or shocks, it is often necessary to consider the development of a stochastic process, which models aging, wearing, damage accumulation, and other changes of the state. In the following, we will call such processes *degradation models*.

The applicability of a purely lifetime-based reliability analysis is limited due to several reasons. Collecting a sample of lifetime data is often expensive. In applications from engineering, observed lifetimes are frequently very long. Furthermore, there are many products for which breakdown causes serious difficulties.

Starting from basic technical considerations, a suitable mathematical model for the degradation measure process $Z(t)$ has to be defined. For instance, the hypothesis of additive degradation accumulation may be used. In this case, we may select the class of homogeneous processes with independent increments as a potentially suitable model class. By applying the level exceeding theory, it is possible to find the type of lifetime distribution, ie, the distribution of the time at which the stochastic process firstly exceeds the degradation level. Then, the parameters of the selected model can be estimated by means of samples related to the degradation measure $Z(t)$ and estimations of the parameters of lifetime distribution can be obtained from the estimated parameters of the model $Z(t)$. There are a lot of papers and books concerning with degradation processes; see, eg, other works.^{1–5} The most classical stochastic processes used for degradation modeling are the Wiener and the Gamma processes. Both processes are processes with independent and stationary increments. The Gamma process is a monotone increasing jump process with Gamma distributed increments, whereas the Wiener process has continuous paths and Gaussian distributed increments. One of the advantages of the Wiener process is that the resulting lifetime distribution can be given explicitly. For more details about degradation processes, we refer the work of Kahle et al⁶ and references given therein.

In the work of Kahle,⁷ a complete optimal preventive maintenance was considered. We assume that the preventive maintenance is *imperfect*, that is, by the maintenance action, the degradation level is reduced with some degree. In this paper, we consider the influence of preventive maintenance to the behavior of the degradation process.

One commonly used class of imperfect maintenance models is the virtual age model introduced by Kijima et al.^{8,9} The virtual age model assumes that an imperfect repair reduces the age of a system by some factor. The concept of virtual age for modeling imperfect maintenance was used by many authors, eg, by Pham and Wang,¹⁰ Lindqvist,¹¹ and Chan and Shaw.¹² These models are based on a reduction of age or a reduction of the failure rate, see, among others, the work of Doyen and Gaudoin.¹³ Using degradation models, there arises the opportunity to consider degradation-based maintenance. In some papers, eg, those by Mercier and Castro¹⁴ and Zhang et al.,¹⁵ imperfect maintenance is reducing the degradation level of a maintained system or the degradation factor. These papers are dealing with the Gamma degradation process. There are only few papers considering the imperfect maintenance in Wiener degradation models, eg, those by Guo et al.¹⁶

Our aim is to consider in more details the influence of reduction of degradation level in Wiener process models. Since for the Wiener process with linear drift the lifetime distribution is given explicitly, the question arises of the connection between degradation-based maintenance and age-based maintenance. The paper is structured as follows. After an introduction to Wiener process and to Kijima-type imperfect repair, in Section 2, possible reductions of the degradation level are described. In Section 3, we analyze the influence of preventive imperfect maintenance to the mean and variance functions of the degradation process. Section 4 contains an investigation of the influence to the resulting first passage time. Furthermore, parameter estimators for the process parameters, as well as for the degree of repair, are developed in Section 5.

1.1 | The degradation process and the resulting lifetime distribution

The choice of the mathematical model for these degradation processes is based on the assumption of an additive accumulation of degradation with constant wear intensity. Regarding every degradation increment as an additive superposition of a large number of small effects, we can assume the degradation process to be normally distributed. Therefore, the degradation measure $Z(t)$ can be described by the following model¹⁷:

$$Z(t) = z_0 + \sigma W(t - t_0) + \mu \cdot (t - t_0), \quad t \geq t_0, \quad (1)$$

with z_0 - constant initial degradation ($z_0 \in \mathbf{R}$),
 t_0 - beginning of the degradation ($t_0 \in \mathbf{R}$),
 μ - drift parameter ($\mu \in \mathbf{R}$),
 σ - variance parameter ($\sigma > 0$),
 $W(t)$ - standard Wiener process on $[0, \infty)$.

We suppose that a failure of a product will occur if the degradation process arrives at a certain boundary degradation level. For a given boundary level h , the lifetime T_h of the product is then determined as the instant at which the degradation process $Z(t)$ reaches the level h for the first time

$$T_h = \inf \{t \geq t_0 : Z(t) \geq h\}. \quad (2)$$

It is well known that for $z_0 < h$, the lifetime T_h follows an inverse Gaussian distribution with the Lebesgue density function

$$f_{T_h}(t) = \frac{h - z_0}{\sqrt{2\pi\sigma^2(t - t_0)^3}} \exp\left(-\frac{(h - z_0 - \mu(t - t_0))^2}{2\sigma^2(t - t_0)}\right) I_{\{t > t_0\}}, \quad (3)$$

where $I_{\{\cdot\}}$ denotes an indicator variable, assuming the value 1 if the relation in brackets is true, and the value 0 otherwise. Obviously, P^{T_h} is the Dirac measure ϵ_{t_0} concentrated at t_0 if $z_0 \geq h$.

Usually, the following alternative parameterization of the inverse Gaussian Lebesgue density function is also used, as example, in the work of Seshadri¹⁸ Chhikara and Folks¹⁹:

$$f(t) = \sqrt{\frac{\lambda}{2\pi t^3}} \exp\left(\frac{-\lambda(t - \mu_1)^2}{2\mu_1^2 t}\right) I_{\{t > 0\}}.$$

Setting

$$\left\{ \begin{array}{l} \lambda = \frac{(h-z_0)^2}{\sigma^2} \\ \mu_1 = \frac{h-z_0}{\mu} \end{array} \right\} \Longleftrightarrow \left\{ \begin{array}{l} \sigma^2 = \frac{(h-z_0)^2}{\lambda} \\ \mu = \frac{h-z_0}{\mu_1} \end{array} \right\},$$

we can see that the two densities are equivalent for $t_0 = 0$. $t_0 \in \mathbf{R}$ is an additional location parameter and we get the three-parameter inverse Gaussian distribution (see the work of Padgett and Wei²⁰ for instance).

Several generalizations of this simple model were given. It is possible to include measurement errors,²¹ or to transform the time scale.²² It is also possible to consider bivariate models.²³ A similar model and its application in medicine are described in the work of Doksum and Normand.²⁴

The advantages of using the Wiener process and its generalizations for describing the degradation process are its simple form (at least for the univariate Wiener process), and, secondly, that a statistical analysis can be carried out from observations at any discrete time points. The parameters of the inverse Gaussian distribution coincide with those of the degradation process.

1.2 | Kijima imperfect preventive maintenance

In Kijima-type^{8,9} models, an item with initial failure rate $\lambda_1(t) = \lambda(t)$ is considered. At some time t_1 (this can be a failure time or a stopping time for preventive maintenance; see the work of Doyen and Gaudoin¹³), the item will be repaired with the degree ξ_1 . By this repair action, the age of the item is decreased to $v_1 = \xi_1 t_1$. The distribution of the time until the next failure has the failure rate $\lambda_2(t) := \lambda(t - t_1 + v_1)$. Let $t_1, t_2, \dots$ be the sequence of maintenance points. The process defined by

$$v(t) := t - t_k + v_k, \quad t_k \leq t < t_{k+1}, \quad n \geq 1,$$

where v_k is the age after the k th repair, is called the *virtual age process*. Kijima has considered two possible types for the reduction of the virtual age:

$$\text{Kijima type I: } v_k = v_{k-1} + \xi_k(t_k - t_{k-1})$$

$$\text{Kijima type II: } v_k = \xi_k(v_{k-1} + (t_k - t_{k-1})).$$

The distribution of the time until the next failure then has failure intensity

$$\lambda_{k+1}(t) = \lambda(t - t_k + v_k). \quad (4)$$

2 | DEGRADATION PROCESSES WITH IMPERFECT PREVENTIVE MAINTENANCE

We consider the degradation process $Z(t)$ that starts at time $t_0 = 0$ in state $z_0 = 0$ and is maintained at $t_1, t_2, \dots$. For simplicity, let the maintenance points have the same distance τ , ie, $t_k = k\tau$. Let the state of the process at first maintenance be $Z(\tau -) = z_1$ and let ξ be the degree of repair.

There are 2 possibilities of an imperfect repair:

- (a) Reduction of degradation level: $z(\tau) = \xi \cdot z_1$.
- (b) It is also possible to reduce the virtual age $v(\tau) = \xi \cdot \tau$ and reset the state to $z(\tau) = \xi \cdot \mu \cdot \tau$ which is the reduced mean state at time τ . Note that, here, it is possible that, after the state correction, the process is worse than before if $Z(\tau -)$ is less than $\xi \mu \tau$.

At the second maintenance point, the state of the process is $Z(2\tau -) = z_2$. Again, there are two possibilities, and for each of them, we can make a reduction of degradation level similar to Kijima type I or type II:

- (a) Reduction of degradation level:

- Kijima type I: $z(2\tau) = \xi \cdot z_1 + \xi(z_2 - z_1) = \xi z_2$,
- Kijima type II: $z(2\tau) = \xi \cdot (\xi \cdot z_1 + (z_2 - z_1)) = \xi^2 \cdot z_1 + \xi(z_2 - z_1)$.

(b) Reduction of virtual age:

- Kijima type I: $z(2\tau) = \xi \cdot \mu \cdot 2\tau$,
- Kijima type II: $z(2\tau) = \xi^2 \cdot \mu \cdot \tau + \xi \mu \cdot (2\tau - \tau) = (\xi^2 + \xi) \cdot \mu \cdot \tau$.

A general expression for $z(k\tau)$ is given by

(a) Reduction of degradation level:

- Kijima type I: $z(k\tau) = \xi z_k$,
- Kijima type II: $z(k\tau) = \sum_{l=1}^k \xi^{k+1-l} (z_l - z_{l-1})$, $z_0 = 0$.

(b) Reduction of virtual age:

- Kijima type I: $z(k\tau) = \xi \cdot \mu \cdot k\tau$,
- Kijima type II: $z(k\tau) = \mu \cdot \tau \cdot \sum_{l=1}^k \xi^l = \mu \cdot \tau \cdot \frac{\xi(1-\xi^k)}{1-\xi}$.

Note that for case (a) (reduction of degradation level), the reduction of the state is random depending on the state just before maintenance action, whereas for case (b) (reduction of virtual age), the new state of the degradation process is fixed. Furthermore, in both cases, the boundary value $\xi = 0$ leads to a renewal of the system. If $\xi = 1$, in case (a), we get a system which is as bad as old, but in case (b) at each maintenance point, the state of the process is adjusted to the mean value of the process.

The behavior of the process under reduction of degradation level (case (a)) with $\xi = 0.6$ and $\tau = 1$ is given in Figure 1. The full line is the original process, whereas the dashed and dotted lines are the processes under state reduction for Kijima type I and type II, respectively.

3 | MEAN AND VARIANCE FUNCTIONS UNDER DEGRADATION REDUCTION

3.1 | Reduction of degradation level

As mentioned before, we restrict ourselves to maintenance point with the same distance τ and to the same degree of repair ξ at each maintenance point. When the process is modified at maintenance times, it has not longer stationary

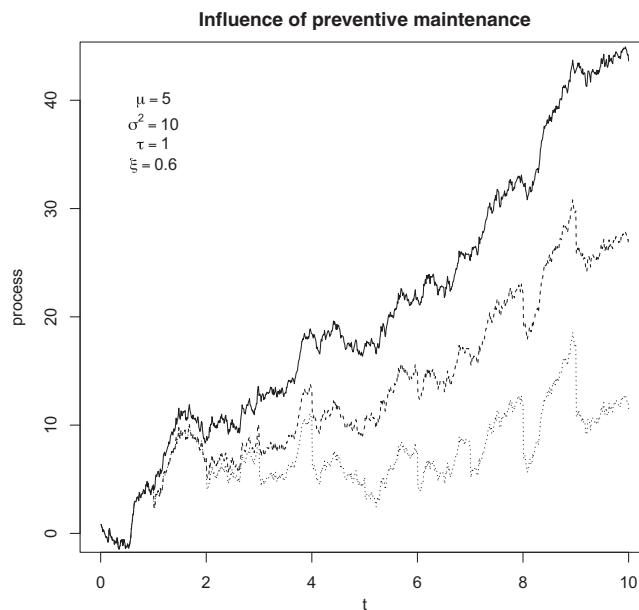


FIGURE 1 The influence of preventive maintenance to the degradation process

increments in general. Since the process has a fixed state after each imperfect repair at maintenance points, it holds that $\{Z(t) - Z(k\tau) | Z(k\tau) = z_k\} \sim N(\mu(t - k\tau), \sigma^2(t - k\tau))$ for $k\tau \leq t < (k + 1)\tau$. Now, we get

- Kijima type I:

For Kijima type I and case (a), the state of the process at time $t, k\tau \leq t < (k + 1)\tau, k = 0, 1, \dots$ is

$$Z(t) = \sum_{l=1}^k \xi (Z(l\tau-) - Z((l-1)\tau)) + (Z(t) - Z(k\tau)),$$

with $Z(0) = 0$. From the Gaussian distribution of increments, we get

$$\begin{aligned} E(Z(t)) &= \xi \mu k\tau + \mu(t - k\tau) \quad \text{and} \\ \text{Var}(Z(t)) &= \xi^2 \sigma^2 k\tau + \sigma^2(t - k\tau). \end{aligned}$$

Since $k\tau \leq t < (k + 1)\tau$, we get boundaries for mean and variance functions

$$\xi \mu t \leq E(Z(t)) < \xi \mu t + \mu \tau, \quad \xi^2 \sigma^2 t \leq \text{Var}(Z(t)) < \xi^2 \sigma^2 t + \sigma^2 \tau.$$

Both functions are increasing in time as before but with a smaller velocity.

- Kijima type II:

For Kijima type II and case (a), the state of the process at time $t, k\tau \leq t < (k + 1)\tau, k = 0, 1, \dots$ is

$$Z(t) = \sum_{l=1}^k \xi^{k+1-l} (Z(l\tau-) - Z((l-1)\tau)) + (Z(t) - Z(k\tau)),$$

with $Z(0) = 0$. Here, we get for expectation and variance functions

$$\begin{aligned} E(Z(t)) &= \sum_{l=1}^k \xi^{k+1-l} E(Z(l\tau) - Z((l-1)\tau)) + E(Z(t) - Z(k\tau)) \\ &= \mu \tau \sum_{l=1}^k \xi^{k+1-l} + \mu(t - k\tau) = \frac{\xi(1 - \xi^k)}{1 - \xi} \mu \tau + \mu(t - k\tau) \quad \text{and} \\ \text{Var}(Z(t)) &= \frac{\xi^2(1 - \xi^{2k})}{1 - \xi^2} \sigma^2 \tau + \sigma^2(t - k\tau). \end{aligned}$$

It can be seen that for $k \rightarrow \infty$ both functions tend to constant values, the process becomes stationary in some sense.

3.2 | Reduction of virtual age

If at each maintenance time $k\tau$ the state of the process is reduced to a fixed value which does not depend on the process development in the past, then the process starts again but with other initial values.

- Kijima type I:

For Kijima type I and case (b) at each maintenance point $k\tau, k = 1, 2, \dots$, the process starts again with initial values $t_0 = k\tau$ and $z_0 = \xi \mu k\tau$. Then, the state of the process at time $t, k\tau \leq t < (k + 1)\tau, k = 0, 1, \dots$ is

$$Z(t) = \xi \mu k\tau + Z(t - k\tau).$$

For mean and variance functions, we get

$$\begin{aligned} E(Z(t)) &= \xi \mu k\tau + \mu(t - k\tau) \quad \text{and} \\ \text{Var}(Z(t)) &= \sigma^2(t - k\tau). \end{aligned}$$

The mean function is the same as for state reduction, but the variance is much smaller.

- Kijima type II:

The difference to Kijima type I is only in the state at new start, z_0 is now $\mu \tau \sum_{l=1}^k \xi^l = \mu \tau \frac{\xi(1 - \xi^k)}{1 - \xi}$ and we get

$$\begin{aligned} E(Z(t)) &= \mu \tau \frac{\xi(1 - \xi^k)}{1 - \xi} + \mu(t - k\tau) \quad \text{and} \\ \text{Var}(Z(t)) &= \sigma^2(t - k\tau). \end{aligned}$$

4 | THE INFLUENCE OF DEGRADATION REDUCTION TO THE DISTRIBUTION OF FAILURE TIME

As mentioned before, there is a direct connection between degradation development and lifetime distribution. This leads to the question about the influence of degradation reduction to the distribution of failure time. On one side, in Kijima imperfect maintenance models, the virtual age of the system is reduced by some degree. On the other side, in our degradation process, we reduce the degradation level. Has it the same influence to the resulting lifetime distribution as the reduction of virtual age? To answer this question, we first consider changes in failure time distribution for Kijima models, then we are asking about a connection to degradation reduction.

4.1 | Kijima preventive maintenance

Let T be a lifetime distribution. We assume that at time points $k \cdot \tau$, $k = 1, 2, \dots$ a preventive maintenance is performed, which reduces the virtual age to v_k . Let $\Lambda_0(t) = \int_0^t \lambda_0(s)ds$ be the cumulative hazard function of a new item and let $\Lambda_k(t)$ be the cumulative hazard function within $[(k-1)\tau, k\tau]$.

Assuming that there are no failures, we get from (4) with $v_0 = 0$

$$\begin{aligned}\Lambda_1(t) &= \Lambda_0(t) \\ \Lambda_2(t) &= \int_0^\tau \lambda_0(s)ds + \int_\tau^t \lambda_0(s - \tau + v_1)ds \\ &= \Lambda_0(\tau) + \Lambda_0(t - \tau + v_1) - \Lambda_0(v_1) \\ &\dots \\ \Lambda_{k+1}(t) &= \sum_{j=0}^{k-1} (\Lambda_0(\tau + v_j) - \Lambda_0(v_j)) + \Lambda_0(t - k\tau + v_k) - \Lambda_0(v_k).\end{aligned}$$

Using the connection between cumulative hazard and density function

$$\begin{aligned}f_{k+1}(t) &= \lambda_{k+1}(t) \exp(-\Lambda_{k+1}(t)) \\ &= \lambda_0(t - k\tau + v_k) \exp\left(-\sum_{j=0}^{k-1} (\Lambda_0(\tau + v_j) - \Lambda_0(v_j)) + \Lambda_0(t - k\tau + v_k) - \Lambda_0(v_k)\right),\end{aligned}$$

we get

$$f_{k+1}(t) = f_0(t - k\tau + v_k) \cdot C(k, \mathbf{v}, \tau)$$

with

$$C(k, \mathbf{v}, \tau) = \frac{\exp\left(-\sum_{j=0}^{k-1} \Lambda_0(\tau + v_j)\right)}{\exp\left(-\sum_{j=1}^k \Lambda_0(v_j)\right)}, \quad \mathbf{v} = \{v_1, \dots, v_k\}.$$

As mentioned before, the resulting lifetime distribution, if degradation follows a Wiener process with drift, is the inverse Gaussian distribution (3), here with $z_0 = t_0 = 0$. In Figure 2, the behavior of the time-to-failure density under Kijima preventive maintenance is plotted for $\mu = 5$, $\sigma^2 = 10$, $h = 40$, $\tau = 4$. Furthermore, it was assumed a Kijima type I preventive maintenance with $\xi_k = \xi = 0.5$. The dotted lines are densities without preventive maintenance, with only one preventive maintenance and so on, the full line is the density after five preventive maintenances.

Furthermore, we can consider the influence of preventive maintenance to the expected lifetime. Let us consider one reduction of age at time τ with $v_1 = \xi\tau$ and let $\tilde{T}_h$ be the new lifetime. Then, it follows

$$E(\tilde{T}_h) = \int_0^\tau t f_0(t)dt + \frac{\bar{F}_0(\tau)}{\bar{F}_0(\xi\tau)} \int_\tau^\infty t f_0(t - \tau + \xi\tau)dt.$$

By partial integration, we get

$$E(\tilde{T}_h) = \int_0^\tau \bar{F}_0(t)dt + \frac{\bar{F}_0(\tau)}{\bar{F}_0(\xi\tau)} \int_\tau^\infty \bar{F}_0(t - \tau + \xi\tau)dt.$$

For distributions with increasing failure rate, it holds

$$\frac{\bar{F}_0(t_1 + s)}{\bar{F}_0(t_1)} \geq \frac{\bar{F}_0(t_2 + s)}{\bar{F}_0(t_2)}, \quad t_1 \leq t_2, \quad s \geq 0.$$

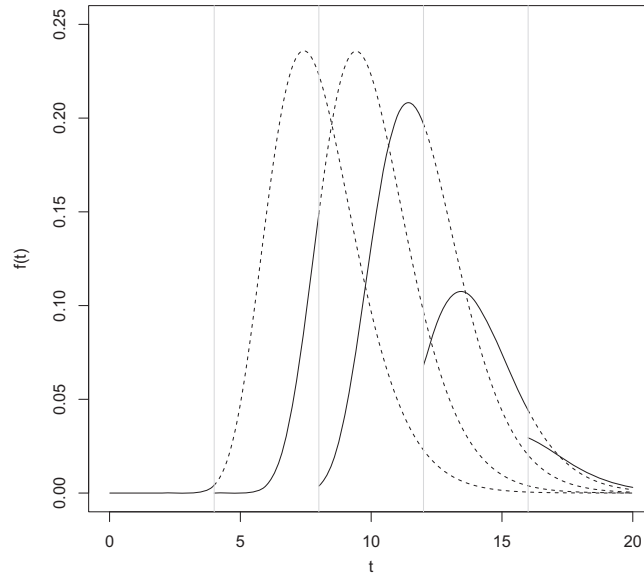


FIGURE 2 The influence of preventive maintenance to time-to-failure distribution

Choosing $s = t - \tau$, $t_1 = \xi\tau$, $t_2 = \tau$, we get

$$\frac{\bar{F}_0(t - \tau + \xi\tau)}{\bar{F}_0(\xi\tau)} \geq \frac{\bar{F}_0(t)}{\bar{F}_0(\tau)}$$

and

$$E(\tilde{T}_h) \geq \int_0^\tau \bar{F}_0(t) dt + \frac{\bar{F}_0(\tau)}{\bar{F}_0(\xi\tau)} \frac{\bar{F}_0(\xi\tau)}{\bar{F}_0(\tau)} \int_{\xi\tau}^\infty \bar{F}_0(t) dt = E(T_h).$$

4.2 | Influence of degradation reduction to lifetime distribution

Now, we investigate the influence of degradation reduction to the resulting life time. For simplicity, we consider only one maintenance time τ . There is no failure up to $t > \tau$ if:

- the process has survived time τ and is in state $z < h$ at τ ,
- the process with new initial state ξz survived time $t - \tau$,

and one has to integrate over all $z \in (-\infty, h)$.

First, we need the density of the conditional distribution of an increment $z < h$ up to time τ **no passing the boundary** h **within** $[0, \tau]$. In the work of Kahle and Lehmann,²⁵ it was found that this density of the conditional distribution that the process is in state $z < h$ at time t given that it was in state $x < h$ at time s is

$$f(s, x, t, z, h) = \frac{1}{\sigma\sqrt{t-s}} \phi\left(\frac{z-x-\mu(t-s)}{\sigma\sqrt{t-s}}\right) \left[1 - \exp\left(-\frac{2(h-x)(h-z)}{\sigma^2(t-s)}\right)\right] I_{\{z \leq h\}},$$

where ϕ is the density of the standard normal distribution. The difference to an unconditional Gaussian increment is the term in square brackets: As closer the difference $(h - x)$ is, as smaller is this term. Some such densities are plotted in Figure 3 for $h = 20$. It can be seen that the area under density becomes smaller with increasing x because, for x close to h , only a small part of all sample paths does not cross the boundary level h .

Secondly, after reaching $z < h$ at time τ , the new failure time distribution is an inverse Gaussian distribution with $z_0 = \xi z$ and $t_0 = \tau$. From the formula of total probability, it follows

$$P(\tilde{T}_h > t) = \int_{-\infty}^h F(0, 0, \tau, z, h) d\bar{F}_{T_h}(t, z),$$

where

- $\tilde{T}_h$ is the new time to failure after one reduction of degradation level at time τ ,

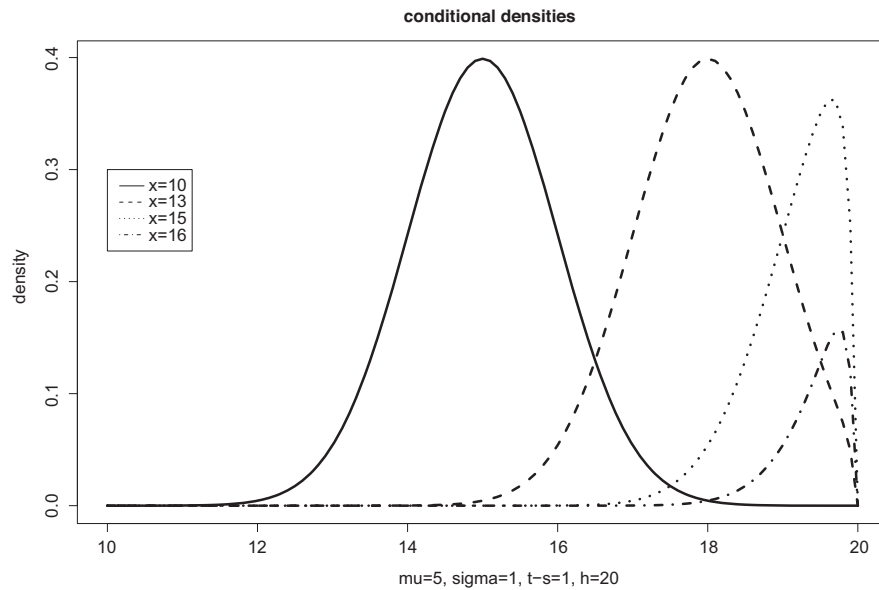


FIGURE 3 Conditional densities for different x

- $F(0, 0, \tau, z, h)$ is the probability that the process starting from state 0 at $t = 0$ is in state $z < h$ at $t = \tau$ without crossing the boundary h in $[0, \tau]$, and
- $\bar{F}_{T_h}(t, z)$ is the probability of surviving the time t . This is the survival function of the inverse Gaussian distribution, but with other parameters: The process is starting now at time $t_0 = \tau$ in state $z_0 = z$ and the new boundary is $\tilde{h} = h - \xi z$. Note that we integrate over all $z < h$, which is included in the parameters of the inverse Gaussian distribution.

Passing to densities, the new failure time distribution after one reduction of the degradation level is given by

$$\tilde{f}_{T_h}(t) = \int_{-\infty}^h f(0, 0, \tau, z, h) f_{T_h}(t, z) dz.$$

The calculation of the integral above is quite difficult. In Figure 4, the results of a simulation are given. For this, 1000 failure times of the Wiener process with one preventive reduction of degradation level are simulated. The parameters are $\mu = 5, \sigma^2 = 3, h = 40, \tau = 6, \xi = 0.5$. In Figure 4, the histogram of these failure times together with a nonparametric density estimation (dotted line) and the density of the inverse Gaussian distribution with age reduction and the same degree (full line) are plotted. The curves coincide very well. The only difference is in the neighborhood of the maintenance

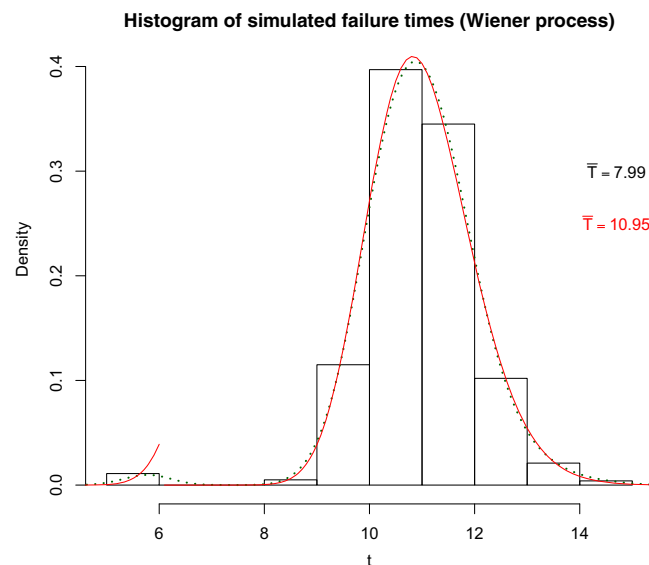


FIGURE 4 Failure times of the Wiener process with reduction of degradation level [Colour figure can be viewed at wileyonlinelibrary.com]

point where the nonparametric density estimate is smooth while the density of the inverse Gaussian distribution with age reduction has a jump at maintenance time. One could well assume that a degradation reduction of the underlying process leads to an age reduction of the resulting lifetime distribution, even with the same degree.

5 | PARAMETER ESTIMATORS

Our last topic is to find parameter estimators for the process parameters μ and σ^2 as well as for the degree of repair ξ . We assume again $t_k = k\tau$ for the maintenance points and $\xi_k = \xi$ for the degree of the k th preventive maintenance action. Furthermore, we assume that we have m maintenance points in one observation of the process (without failure, ie, $h \rightarrow \infty$).

At time $k \cdot \tau, k = 1, \dots, m$, we observe the state of the process z_k and reduce the degradation level to $\xi \cdot z_k$, which is a Kijima type II reduction. Furthermore, we need the assumption that the state of the process after state reduction (and therefore the degree of maintenance) is not observable. This can be the case if degradation is measured by some latent variable, as example, the loss of oil. Using the property of independent increments between maintenance points, we get the following likelihood function for the observation of one process:

$$L = \prod_{j=1}^m \frac{1}{\sqrt{\sigma^2 \tau}} \varphi \left(\frac{z_j - \xi z_{j-1} - \mu \tau}{\sqrt{\sigma^2 \tau}} \right),$$

where φ is the density function of the standard normal distribution.

Taking the derivatives of the likelihood function and solving the system of likelihood equations, we obtain the following estimators:

$$\begin{aligned} \hat{\xi} &= \frac{\sum_{j=1}^m z_j \sum_{j=1}^{m-1} z_j - m \sum_{j=1}^{m-1} z_j z_{j+1}}{\left(\sum_{j=1}^{m-1} z_j \right)^2 - m \sum_{j=1}^{m-1} z_j^2} \\ \hat{\mu} &= \frac{1}{m\tau} \left(\sum_{j=1}^m z_j - \hat{\xi} \sum_{j=1}^{m-1} z_j \right) \\ \hat{\sigma}^2 &= \frac{1}{m\tau} \sum_{j=1}^m (z_j - \hat{\xi} z_{j-1})^2 - \hat{\mu}^2 \tau. \end{aligned}$$

It is easy to generalize this way to arbitrary distances between maintenance times and to the observation of n independent processes. Note that the number of observations and the observation points may be different in each realization.

The following small simulation study shows the goodness of estimators. In the following table, the mean value and standard deviation of parameter estimates from 1000 simulated samples with $\xi = 0.5, \mu = 5, \sigma = 1.5$ are given.

m	ξ	μ	σ
20	0.456 (0.121)	5.403 (1.144)	1.410 (0.237)
40	0.472 (0.098)	5.253 (0.967)	1.460 (0.165)
100	0.482 (0.072)	5.174 (0.720)	1.481 (0.102)
500	0.495 (0.037)	5.050 (0.376)	1.496 (0.047)

The estimators for ξ and μ are indirect proportional each to another: ξ is slight underestimated and μ is slight overestimated. This effect disappears for a large number of observations.

6 | CONCLUSION

In the paper, the Wiener degradation model with imperfect preventive maintenance was considered. For a simple maintenance model (equidistant maintenance points and a constant degree of maintenance), it was analyzed the influence of maintenance to the behavior of the degradation process and to the resulting lifetime distribution. A connection between age reduction in classical lifetime models and degradation reduction of the underlying degradation process was established. In future works, more complicated models can be considered, as example, a random degree of maintenance. Furthermore, a condition-based maintenance could be considered where the degree of repair depends on the measured

state of the process at inspection/maintenance points. Moreover, there are many questions concerning optimal maintenance: how to choose maintenance points (not necessary equidistant) and degrees of maintenance to minimize a given cost function.

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